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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 4.2 Stochastic Gradient Descent (SDG) for the MNIST data set
5  Machine Learning for Engineering Problem Solving
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7  """
8
9  import IPython as IP
10 IP.get_ipython().run_line_magic('reset', '-sf')
11
12 import numpy as np
13 import scipy as sp
14 import matplotlib as mpl
15 import matplotlib.pyplot as plt
16 import sklearn as sk
17 from sklearn import linear_model
18 from sklearn import datasets
19
20 cc = plt.rcParams['axes.prop_cycle'].by_key()['color']
21 plt.close('all')
22
23 %% Load your data
24
25 # Fetch the MNIST dataset from openml
26 mnist = sk.datasets.fetch_openml('mnist_784', as_frame=False, parser='auto')
27 X = mnist['data'] # load the data
28 Y = np.asarray(mnist['target'], dtype=int) # load the target
29
30 # Split the data set up into a training and testing data set
31 X_train = X[0:60000,:]
32 X_test = X[60000:,:]
33 Y_train = Y[0:60000]
34 Y_test = Y[60000:]
35
36 %% Train a Stochastic Gradient Descent classifier
37
38 # Extract a subset for our "5-detector".
39 Y_train_5 = (Y_train == 5)
40 Y_test_5 = (Y_test == 5)
41
42 # build and train the classifier
43 sgd_clf = sk.linear_model.SGDClassifier()
44 sgd_clf.fit(X_train, Y_train_5)
45
46 # get a digit from the dataset to test the classifier on
47 digit_id = 35 # An OK 5
48 # digit_id = 0 # An odd 5
49 # digit_id = 100 # A bad 5
50 test_digit = X[digit_id,:]
51 digit_resaped = np.reshape(test_digit, (28,28))
52
53 # plot an image of the random pixel you picked above.
54 plt.figure()
55 plt.imshow(digit_resaped, cmap = mpl.cm.binary, interpolation="nearest")
56 plt.title('A "'+str(Y[digit_id])+'" digit from the MNIST dataset')
57 plt.xlabel('pixel column number')
58 plt.ylabel('pixel row number')
59 plt.savefig('MNIST_digit')
60
61
62 # we can now test this for the "5" that we plotted earlier.
63 print(sgd_clf.predict([test_digit])) # a True case
64 print(sgd_clf.predict([X[2,:]])) # a False case
65
66
67

```